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Can LLM Agents Learn by Watching Tutorial Videos?

Tutorial Agentic Learning Turns Video Procedures into Frozen-LLM Memory with No Weight Updates

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Tutorial Agentic Learning (TAL) enables a frozen large language model to acquire procedural knowledge from instructional videos at inference time, without any weight updates. Tested on VidLearn—a new multimodal benchmark of 72 chess tutorial videos spanning 12 openings—TAL performance scales with the number of videos consumed, demonstrating genuine procedural learning from visual and language content.

AT A GLANCE

Problem: LLM agents require expensive retraining to acquire new procedural skills, blocking continual learning at deployment.

Why it matters: Tutorial videos are abundant and contain rich procedural knowledge; efficiently converting them to agent capability would eliminate per-skill fine-tuning cycles.

Method: Videos are segmented into timestamped procedural entries and stored in a retrieval-augmented memory bank; at inference time, top-k entries provide in-context guidance to a frozen LLM.

Result: TAL outperforms the same LLM without video context; accuracy scales with number of videos ingested on the VidLearn benchmark.

Evidence: VidLearn: 72 chess tutorial videos, 12 openings, multimodal QA; dense retrieval ablations over retrieval count k.

The Big Picture

The boundary between training and inference has been treated as fixed in standard machine learning: an agent learns during training and exploits what it learned during deployment. But in real-world settings, new procedural knowledge becomes available continuously—software tutorials, instructional videos, how-to guides—and re-initiating a training run for each new skill is impractical.

Tutorial Agentic Learning (TAL) reframes this problem: rather than modifying weights, it treats instructional videos as a persistent, queryable memory that augments a frozen agent at inference time. The approach is inspired by retrieval-augmented generation (RAG) for factual QA, but extends the paradigm to procedural, visually grounded knowledge embedded in sequential video content.

Why Existing Approaches Fall Short

Fine-tuning on instructional data requires labeled examples and gradient computation, is prone to catastrophic forgetting of previously learned skills, and must be repeated for each new skill domain.

In-context learning from text summaries loses the visual content of videos and fails to capture the sequential, state-action structure of procedural knowledge, particularly in domains like chess or software manipulation where board/screen state is essential.

Vision-language models applied to full video at inference time are computationally prohibitive: processing 72 tutorial videos at every query step would require multiple full video passes per query.

Core Mechanism

TAL operates in two phases. During **offline indexing**, each video is segmented into timestamped steps using a vision-language model (VLM). Each step is converted into a structured procedural entry:

$$e_i = (\text{state}_i,\ \text{action}_i,\ \text{rationale}_i)$$

where state_i is a visual embedding of the board/screen, action_i is the described move or operation, and rationale_i is the narrated explanation. All entries are stored in a FAISS-indexed memory bank.

During **inference-time retrieval**, given a query q, the top-k entries are retrieved via dense similarity search:

$$\mathcal{E}^* = \arg\top_k \{\cos(\phi(q),\ \phi(e_i))\}_{i=1}^{N}$$

where $\phi(\cdot)$ is a frozen CLIP encoder. The retrieved entries are formatted as structured in-context examples and provided to the frozen LLM for final answer generation.

The VidLearn Benchmark

VidLearn is a new multimodal QA benchmark constructed from 72 publicly available chess tutorial videos spanning 12 chess openings (Sicilian Defense, King's Indian, French Defense, and 9 others). Each video is annotated with key board positions, move-by-move explanations, and multiple-choice and free-form questions. The benchmark tests whether an agent can correctly identify optimal continuations from visually presented board states, drawing on knowledge explicitly taught in the videos.

The 12 openings are partitioned into train (8) and test (4) splits to evaluate cross-opening transfer.

Learning or Inference Procedure

Offline (once per video set):

1. Segment each video into T_v timestamped steps using a VLM.
2. Embed each step as a (state, action, rationale) triple.
3. Encode each triple with CLIP and add to the FAISS index.

Inference (per query):

1. Encode the query board image and question text with CLIP.
2. Retrieve top-k entries from the memory bank.
3. Format as in-context examples and pass to the frozen LLM.
4. Return the LLM's generated answer.

No gradient computation occurs at any stage of inference.

What the Guarantee Says

TAL provides no formal learning-theoretic guarantee. The empirical claim is a scaling law: accuracy on VidLearn test questions increases as the number of ingested training-opening videos grows, leveling off around 50 videos per opening. This suggests that the memory bank successfully captures the unique procedural patterns in a given opening up to a saturation point. No guarantee is provided for out-of-distribution openings, though cross-opening transfer is observed for structurally related sub-variants.

Experimental Findings

- **Accuracy vs. baseline:** TAL with 72 videos achieves substantially higher accuracy than the same LLM without video context on VidLearn multiple-choice questions.
- **Scaling:** Accuracy increases monotonically with the number of ingested videos, with diminishing returns above 50 videos per opening.
- **Cross-opening transfer:** Positive transfer is observed for Sicilian Defense sub-variants, with 4.1% accuracy improvement on unseen variations when related-opening videos are present.
- **Retrieval ablation:** Performance improves with k up to $k = 5$, then plateaus, suggesting diminishing returns from additional in-context examples beyond 5.

Ablations and Interpretation

Without visual state embeddings: Replacing CLIP visual embeddings with text-only embeddings of move descriptions reduces accuracy by 7.3%, confirming that the board-state visual signal is essential for precise retrieval.

Without the rationale field: Removing the rationale from in-context entries reduces accuracy by 4.9%, indicating that the LLM uses explanatory text from videos to reason about move motivations.

Approximate vs. exact search: Using exact L2 search instead of approximate FAISS search improves accuracy by less than 0.5%, confirming that approximate retrieval is sufficient and scalable.

Impact of VLM quality for segmentation: Replacing the full VLM with a smaller captioning model for step segmentation reduces accuracy by 6.1%, highlighting that the quality of initial video parsing matters.

Reference

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